

1. Writing SELECT Statements to Fetch Data

The SELECT statement is used to retrieve data from one or more tables.

✂ Syntax:

```
sql
```

CopyEdit

```
SELECT column1, column2, ... FROM table_name;
```

✂ Example:

```
sql
```

CopyEdit

```
SELECT name, age FROM students;
```

-

You can use `*` to select all columns:

```
sql
```

CopyEdit

```
SELECT * FROM students;
```

-

You can use **column aliases** for better readability:

```
sql
```

CopyEdit

```
SELECT name AS student_name, age AS student_age FROM students;
```

❏ 2. Using WHERE Clause for Filtering Data

The WHERE clause filters rows based on specific conditions.

❏ Syntax:

```
sql
```

CopyEdit

```
SELECT column1, column2 FROM table_name WHERE condition;
```

❏ Operators:

-

= : Equal

-

!= or <> : Not equal

-

<, >, <=, >=

-

BETWEEN, IN, LIKE, IS NULL, IS NOT NULL

☒ Example:

sql

CopyEdit

```
SELECT name FROM students WHERE age > 18;
```

sql

CopyEdit

```
SELECT * FROM students WHERE name LIKE 'A%'; -- Names starting with A
```

❏ 3. Sorting Data Using ORDER BY

The ORDER BY clause is used to sort the result set in ascending (default) or descending order.

❏ Syntax:

sql

CopyEdit

```
SELECT column1, column2 FROM table_name ORDER BY column1 [ASC|DESC];
```

☒ Example:

sql

CopyEdit

```
SELECT name, age FROM students ORDER BY age DESC;
```

-

Sort by multiple columns:

sql

CopyEdit

```
SELECT * FROM students ORDER BY course ASC, age DESC;
```

❏ 4. Basic JOIN Operations to Combine Data

JOIN combines rows from two or more tables based on a related column.

Types of Joins:

- **INNER JOIN:** Only matching rows
- **LEFT JOIN:** All rows from left, matched rows from right
- **RIGHT JOIN:** All rows from right, matched from left
- **FULL JOIN:** All matched and unmatched rows (not in MySQL)

❏ Syntax (INNER JOIN):

sql

CopyEdit

```
SELECT table1.column1, table2.column2 FROM table1 INNER JOIN table2 ON table1.common_column = table2.common_column;
```

📄 **Example:**

sql

CopyEdit

```
SELECT students.name, courses.course_name FROM students INNER JOIN courses ON students.course_id = courses.course_id;
```

📄 **40 MCQs on SELECT, WHERE, ORDER BY, and JOIN**

☒ Easy Level (10 Questions)

Q1. Which SQL keyword is used to retrieve data from a table?

- A. GET
- B. FETCH
- C. SELECT
- D. PULL

☒ **Answer:** C

☒ **Explanation:** SELECT is used to retrieve data from tables.

Q2. What does SELECT * FROM employees; do?

- A. Deletes all employees
- B. Updates all data
- C. Selects all columns and rows
- D. Selects only employee IDs

☒ **Answer:** C

☒ **Explanation:** * selects all columns from the employees table.

Q3. Which clause filters rows?

- A. FILTER
- B. ORDER BY
- C. GROUP BY
- D. WHERE

☒ **Answer:** D

❏ **Explanation:** WHERE filters data based on conditions.

Q4. Which clause is used to sort results?

- A. SORT BY
- B. ARRANGE BY
- C. ORDER BY
- D. GROUP BY

❏ **Answer:** C

❏ **Explanation:** ORDER BY sorts results in ascending or descending order.

Q5. What is the default order for ORDER BY?

- A. DESC
- B. RANDOM
- C. NONE
- D. ASC

❏ **Answer:** D

❏ **Explanation:** ASC (ascending) is the default.

Q6. Which operator checks if a value exists in a set?

- A. LIKE
- B. IN
- C. BETWEEN

D. IS

☒ **Answer:** B

☒ **Explanation:** IN checks if a value exists in a list.

Q7. Which symbol represents "not equal" in SQL?

A. ==

B. <>

C. !=

D. Both B and C

☒ **Answer:** D

☒ **Explanation:** SQL supports both <> and != for "not equal".

Q8. LIKE 'A%' matches:

A. Names ending with A

B. Names containing A

C. Names starting with A

D. Names only with A

☒ **Answer:** C

☒ **Explanation:** % after A matches any string starting with A.

Q9. What is INNER JOIN used for?

A. To delete rows

- B. To return unmatched data
- C. To return matching rows from two tables
- D. To create tables

☒ **Answer:** C

☒ **Explanation:** INNER JOIN returns matching rows from both tables.

Q10. Which clause is used after WHERE for sorting?

- A. GROUP BY
- B. ORDER BY
- C. HAVING
- D. COUNT

☒ **Answer:** B

☒ **Explanation:** ORDER BY sorts filtered results.

☒ Medium Level (20 Questions)

Q11. Which SQL statement is valid?

sql

CopyEdit

SELECT name, age FROM users;

- A. Valid
- B. Invalid
- C. Updates rows
- D. Creates table

☒ **Answer:** A

☒ **Explanation:** It selects name and age from users.

Q12. What does WHERE age BETWEEN 20 AND 30 do?

- A. Filters age exactly 20 and 30
- B. Filters age below 20
- C. Filters age from 20 to 30 inclusive
- D. Filters age above 30

☒ **Answer:** C

☒ **Explanation:** BETWEEN includes both endpoints.

Q13. Which query returns names of students aged above 18?

sql

CopyEdit

SELECT name FROM students WHERE age > 18;

- A. Correct
- B. Incorrect syntax
- C. Only works for NULL values
- D. Must use HAVING

☒ **Answer:** A

☒ **Explanation:** This is the correct syntax for filtering.

Q14. Which query sorts data in descending order of marks?

sql

CopyEdit

SELECT * FROM result ORDER BY marks DESC;

- A. Wrong
- B. Works correctly
- C. Sorts randomly
- D. Sorts in ascending order

☒ **Answer:** B

☒ **Explanation:** DESC is used for descending sort.

Q15. Which clause returns only rows where the condition is true?

- A. SELECT
- B. FROM
- C. WHERE
- D. GROUP BY

☒ **Answer:** C

☒ **Explanation:** WHERE applies conditions.

Q16. What does JOIN require?

- A. Same table
- B. Common column
- C. Foreign key
- D. Different databases

☒ **Answer:** B

☒ **Explanation:** JOINS are done based on a common column (often a key).

Q17. LEFT JOIN returns:

- A. Only matching rows
- B. Only unmatched right rows
- C. All left rows, matched from right
- D. Right rows only

☒ **Answer:** C

☒ **Explanation:** LEFT JOIN includes all left-side rows.

Q18. Which SQL will get all customers older than 25 and sorted by name?

sql

CopyEdit

SELECT name FROM customers WHERE age > 25 ORDER BY name;

- A. Correct
- B. Needs HAVING

- C. Incorrect ORDER BY
- D. Should use LIMIT

☒ **Answer:** A

☒ **Explanation:** Correct format to filter and sort.

Q19. INNER JOIN requires:

- A. All columns to match
- B. Unique data
- C. Matched values in joined columns
- D. Same primary key

☒ **Answer:** C

☒ **Explanation:** It needs at least one matching value.

Q20. Which keyword is used to alias a column?

- A. USING
- B. ALIAS
- C. AS
- D. JOIN

☒ **Answer:** C

🗒 Medium Level (20 Questions)

Q11. Which SQL statement is valid?

sql

CopyEdit

SELECT name, age FROM users;

- A. Valid
- B. Invalid
- C. Updates rows
- D. Creates table

🗒 **Answer:** A

🗒 **Explanation:** It selects name and age from users.

Q12. What does WHERE age BETWEEN 20 AND 30 do?

- A. Filters age exactly 20 and 30
- B. Filters age below 20
- C. Filters age from 20 to 30 inclusive
- D. Filters age above 30

☒ **Answer:** C

☒ **Explanation:** BETWEEN includes both endpoints.

Q13. Which query returns names of students aged above 18?

sql

CopyEdit

SELECT name FROM students WHERE age > 18;

- A. Correct
- B. Incorrect syntax
- C. Only works for NULL values
- D. Must use HAVING

☒ **Answer:** A

☒ **Explanation:** This is the correct syntax for filtering.

Q14. Which query sorts data in descending order of marks?

sql

CopyEdit

SELECT * FROM result ORDER BY marks DESC;

- A. Wrong
- B. Works correctly
- C. Sorts randomly
- D. Sorts in ascending order

☒ **Answer:** B

☒ **Explanation:** DESC is used for descending sort.

Q15. Which clause returns only rows where the condition is true?

- A. SELECT
- B. FROM

- C. WHERE
- D. GROUP BY

☒ **Answer:** C

☒ **Explanation:** WHERE applies conditions.

Q16. What does JOIN require?

- A. Same table
- B. Common column
- C. Foreign key
- D. Different databases

☒ **Answer:** B

☒ **Explanation:** JOINS are done based on a common column (often a key).

Q17. LEFT JOIN returns:

- A. Only matching rows
- B. Only unmatched right rows
- C. All left rows, matched from right
- D. Right rows only

☒ **Answer:** C

☒ **Explanation:** LEFT JOIN includes all left-side rows.

Q18. Which SQL will get all customers older than 25 and sorted by name?

sql

CopyEdit

SELECT name FROM customers WHERE age > 25 ORDER BY name;

- A. Correct
- B. Needs HAVING
- C. Incorrect ORDER BY
- D. Should use LIMIT

☒ **Answer:** A

☒ **Explanation:** Correct format to filter and sort.

Q19. INNER JOIN requires:

- A. All columns to match
- B. Unique data
- C. Matched values in joined columns
- D. Same primary key

☒ **Answer:** C

☒ **Explanation:** It needs at least one matching value.

Q20. Which keyword is used to alias a column?

A. USING

B. ALIAS

C. AS

D. JOIN

☒ **Answer:** C

☒ **Explanation:** AS is used to rename columns.

(Remaining 10 medium + 10 hard questions in next message...)

You said:

yes

ChatGPT said:

Continuing with the remaining **MCQs** for Session 2 & 3 topics:

☒ Medium Level (Continued: Q21–Q30)

Q21. What does this query return?

sql

CopyEdit

```
SELECT * FROM employees WHERE department = 'HR' ORDER BY salary DESC;
```

- A. All HR employees sorted by salary ascending
- B. All employees with salary DESC
- C. All HR employees sorted by salary descending
- D. Syntax error

☒ **Answer:** C

☒ **Explanation:** It filters rows from HR and sorts them in descending order of salary.

Q22. The result of SELECT name FROM users WHERE name LIKE '__a%'; will show:

- A. Names with two letters
- B. Names with 'a' at third position
- C. Names starting with a
- D. Names ending with a

☒ **Answer:** B

☒ **Explanation:** __a% means third character is 'a'.

Q23. Which SQL clause limits the number of rows returned?

- A. WHERE
- B. ORDER BY
- C. LIMIT
- D. TOP

☒ **Answer:** C (MySQL/PostgreSQL)

☒ **Explanation:** LIMIT restricts the result set to specified number of rows.

Q24. What does this join do?

sql

CopyEdit

```
SELECT s.name, c.course_name FROM students s JOIN courses c ON s.course_id = c.id;
```

- A. Shows all students only
- B. Shows course names only
- C. Joins students and courses where IDs match
- D. Joins without any condition

☒ **Answer:** C

☒ **Explanation:** The join returns records with matching course_id.

Q25. Which is more efficient when filtering rows?

- A. WHERE
- B. ORDER BY
- C. SELECT
- D. DISTINCT

☒ **Answer:** A

☒ **Explanation:** WHERE reduces row count early in execution.

Q26. ORDER BY salary DESC, age ASC means:

- A. Salary and age both descending
- B. Sorted only by age
- C. Salary descending, age ascending
- D. Age ascending then salary

☒ **Answer:** C

☒ **Explanation:** It sorts first by salary (desc), then age (asc) if salary ties.

Q27. What does % symbol do in LIKE condition?

- A. Replaces one character
- B. Matches a word
- C. Matches any string of characters
- D. No function

☒ **Answer:** C

☒ **Explanation:** % matches zero or more characters.

Q28. Which of the following is valid JOIN syntax?

sql

CopyEdit

SELECT a.id, b.salary FROM emp a INNER JOIN salary b ON a.id = b.emp_id;

- A. Invalid
- B. Missing WHERE
- C. Valid
- D. No ON keyword needed

☒ **Answer:** C

☒ **Explanation:** Correct INNER JOIN format with alias.

Q29. Which clause executes first logically?

- A. SELECT
- B. ORDER BY
- C. WHERE
- D. FROM

☒ **Answer:** D

☒ **Explanation:** SQL logical processing order starts with FROM.

Q30. What does WHERE salary IS NOT NULL do?

- A. Selects rows with NULL salary
- B. Filters only rows with salary values
- C. Removes all rows

D. Shows NULLs only

☒ **Answer:** B

☒ **Explanation:** Filters out NULLs from the result.

☒ Hard Level (10 Questions)

Q31. Given two tables students(id, name, course_id) and courses(id, name), how do you fetch student names with their course names?

A.

sql

CopyEdit

```
SELECT s.name, c.name FROM students s JOIN courses c ON s.id = c.id;
```

B.

sql

CopyEdit

```
SELECT s.name, c.name FROM students s JOIN courses c ON s.course_id = c.id;
```

C.

sql

CopyEdit

```
SELECT name FROM students;
```

D.

sql

CopyEdit

SELECT name FROM courses;

☒ **Answer:** B

☒ **Explanation:** Join using s.course_id = c.id gives proper relation.

Q32. What happens if you use JOIN without ON clause?

- A. SQL error
- B. Cross join (cartesian product)
- C. Only left rows shown
- D. Filters nothing

☒ **Answer:** B

☒ **Explanation:** Without ON, a cartesian product is returned.

Q33. Which statement is TRUE about INNER JOIN and WHERE?

- A. WHERE replaces JOIN
- B. Both must be used
- C. Filtering can happen after JOIN using WHERE
- D. They are unrelated

☒ **Answer:** C

🔍 **Explanation:** Use JOIN for linking, and WHERE for filtering after join.

Q34. What does this return?

sql

CopyEdit

```
SELECT * FROM A LEFT JOIN B ON A.id = B.id WHERE B.id IS NULL;
```

- A. All matching rows
- B. All unmatched rows from A
- C. All rows from B
- D. Inner join

🔍 **Answer:** B

🔍 **Explanation:** This returns unmatched rows from A (anti-join pattern).

Q35. Which statement is best for performance with large datasets?

- A. Avoid WHERE

- B. Use ORDER BY on all columns
- C. Use WHERE with indexes
- D. Always use *

☒ **Answer:** C

☒ **Explanation:** Indexed WHERE conditions reduce scan time.

Q36. If you need to find duplicate names in a table, which clause do you use?

- A. GROUP BY and HAVING
- B. ORDER BY only
- C. WHERE only
- D. JOIN

☒ **Answer:** A

☒ **Explanation:** Use GROUP BY name HAVING COUNT(name) > 1.

Q37. Which returns students not enrolled in any course?

sql

CopyEdit

SELECT s.name FROM students s LEFT JOIN enrollments e ON s.id = e.student_id WHERE e.course_id IS NULL;

- A. Enrolled students
- B. All students
- C. Unenrolled students
- D. Only teachers

☒ **Answer:** C

☒ **Explanation:** LEFT JOIN ... WHERE IS NULL filters unmatched.

Q38. When is USING() used in JOINS?

- A. When column names differ
- B. When joining on multiple keys
- C. When joining tables with same column names
- D. For non-relational DBs

☒ **Answer:** C

☒ **Explanation:** USING(column) can be used when both tables share column name.

Q39. What is the result of SELECT name FROM users ORDER BY 1;?

- A. Ordered by the first row
- B. Ordered by the first column
- C. Orders randomly
- D. Error

☒ **Answer:** B

☒ **Explanation:** ORDER BY 1 means sort by the first selected column.

Q40. Why is SELECT * discouraged in production?

- A. It's slow
- B. Fetches unnecessary data
- C. Hard to optimize
- D. All of the above

☒ **Answer:** D

☒ **Explanation:** SELECT * affects performance, readability, and indexing.

☒ **Explanation:** AS is used to rename columns.